

A Survey of Neuro-Symbolic Approaches in AI

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The development of Large Language Models (LLMs) has been a key focus in Artificial Intelligence (AI) research over the past decade due to extensive corporate funding and competition. Although LLMs are heavily leveraged today for assistance and even automation for various tasks, their utility is limited by a tendency to *hallucinate* — making blatant errors for tasks involving reasoning. Predominantly comprised of neural networks (NNs), LLMs are inherently incapable of performing true logical reasoning, only attempting to do so via means of estimation. While Chain-of-Thought Prompting can drastically improve LLM performance by simulating the thought process, it is not only costly token-wise, but also does not truly resolve the probabilistic issue of LLMs. On the other hand, symbolic AI is an older paradigm utilizing concrete, symbolic representations of problems, states and actions, and thus can be said to perform logical reasoning. Recent studies have proposed hybrid approaches consisting of *connectionist* (NN-based) and symbolic AI — known collectively as neuro-symbolic AI — aiming to bring out the best of both worlds. In this survey, we explore multiple approaches of neuro-symbolic AI: Reasoning, Learning and Metacognition, where we mention significant works in the field and analyze their key factors for success. Finally, we conclude by highlighting promising areas for further development.

1 Introduction

The term "AI", an abbreviation of artificial intelligence, is frequently used today to refer to commercial LLMs (e.g. *ChatGPT*, *Deepseek*, etc.), due to rapid industrialization over the past decade that has significantly influenced our very way of life. Yet, AI is much more than just LLMs. Its underlying concept revolves around an algorithmic reasoning system — a list of steps that can simulate the decisions made by intelligent life, tracing back to the 1st millennium BC. It was not until the birth of computers in the 1900s that such reasoning could finally be carried out on a meaningfully large scale, giving rise to the study of a class of algorithms collectively known as AI. This field of study has branched into multiple paradigms, such as machine learning, evolutionary algorithms, and Bayesian AI, to name a few. In this paper, we explore both symbolic and connectionist AI, and the integration of both systems, known loosely as neuro-symbolic AI.

1.1 A History of Two Approaches

Symbolic AI was the dominating paradigm in the mid-1900s. Its defining trait is the usage of symbols to represent concrete objects and variables, and a collection of symbols to represent a state/statement [19]. Typically, a rules-based program then advances or generates successors to the current state, possibly representing: **I** actions that could be performed in the current state, e.g., move piece left/right/up/down; **II** ways to reformulate the current statement(s), e.g. implication law in Boolean logic; note that this list is not exhaustive. A *Checkers* program developed by computer scientist Arthur Samuel in 1952, which first utilized *minimax with alpha-beta pruning*, attracted much attention towards the symbolic approach due to its perceived effectiveness for a game with a complex structure. Successive advancements such as the *Logic Theorist* in 1956, which formed the basis of formal systems, and the programming language *Lisp* in 1960, created with the intent to accommodate symbolic expressions, would further boost the popularity of this approach. However, symbolic AI had some major shortcomings: a set of symbols and rules had to be manually feature engineered for every specific task, and brute-force search methods meant the state space would grow exponentially with the complexity of the problem. In other words, this approach had very poor scalability, and periods of hype towards implementing a universal model were soon followed by the bleak realization of impossibility, leading to events known as the 1st and 2nd AI winters, where funding and research towards AI were cut short as a result.

The connectionist approach, utilizing artificial neural networks (NNs), was also in development at the same time in mid-1900s, though to a much lesser extent. Drawing inspiration from the biological neuron, and how connections between neurons gave rise to intelligence, a model called the *perceptron* was established, utilizing numerical values known as weights to form connections between intermediate and output layers, and trained iteratively through a primitive backpropagation algorithm. It was also said to be a black-box model, whose internal state was not human-interpretable, unlike most symbolic approaches that utilized concrete representations. The early *perceptron* had issues categorizing nonlinear data, lacking in both size and the presence of activation functions necessary to serve as a universal function approximator. Though its potential was not apparent at the time, multiple breakthroughs over the next few decades eventually refined it into the feedforward neural network, a key component of deep learning (DL) and modern architectures such as LLMs.

1.2 Emergence of Neuro-Symbolic AI

In the present day, the majority of AI research focus areas, such as multimodal AI, reinforcement learning (RL), computer vision, and natural language processing revolve around the connectionist approach due to the advantage that NNs offer in terms of scalability, parallelizability and learning from nuanced data. Yet originating from the very issues of said approach, recurring themes of problems faced across various fields have surfaced, making the development of solutions or mitigation strategies a significant area of study. Such issues include: **I** the validation of a model’s internal state, making sure there is no unintended gaps or defects encoded within its parameters which could lead to undesired outputs, or make it susceptible to adversarial attacks; **II** getting neural architectures to perform reasoning accurately, even though they are prone to the phenomenon known as hallucinations, where wrong or illogical outputs are generated probabilistically; **III** ensuring knowledge is accurately learned and retained, instead of only picking up fragments of information, or knowledge disappearing after subsequent iterations of parameter updates. To address such issues, some recent works have integrated symbolic methods into modern-day connectionist systems, resulting in a broad class of approaches known as neuro-symbolic AI.

2 Definition

Across different literatures, neuro-symbolic AI is defined rather loosely and contrastingly, which we will first clarify. We base our categorization on a system proposed by H. Kautz [11, 25], that defines neuro-symbolic AI as techniques belonging to the following categories:

1. symbolic Neuro symbolic: A model with symbolic input and output data, but intermediate data are represented as vectors that are processed by NNs. An example would be Transformer-based language models, which encodes tokens (symbols) into their respective embeddings and decodes the output of the NN back into readable tokens.

2. Symbolic[Neuro]: A primarily symbolic model that calls a neural subroutine for feedback. Examples include *AlphaGo* and *AlphaZero* that have the neural model serve as a heuristic function to guide a symbolic Monte-Carlo tree search algorithm [28, 29].

3. Neuro | Symbolic: A system that uses a neural model to convert subsymbolic (numerical, image, audio, etc.) inputs to symbolic data which is transferred into a symbolic model. In a sense, both components have equal importance in performing a task. In the *Neuro-Symbolic Concept Learner* [16], the symbolic unit also serves to train the neural network for visual inputs.

4. Neuro: Symbolic $\rightarrow$ Neuro: Approaches that train a neural model to generate symbolic output from symbolic input, by using a symbolic system to generate training data, such as *Deep Learning for Symbolic Mathematics* [13].

5. Neuro_{Symbolic}: Neural architectures that base their structure on symbolic rule representations, enabling rules to be learned through training. Examples are *Tensor Product Representations* and *Logic Tensor Networks* [27, 30].

6. Neuro[Symbolic]: A primarily neural model that may call the symbolic reasoning unit to perform reasoning tasks that already have well-established rules.

As it can be argued that methods that rely on purely neural architectures during the inference phase, if applicable, no longer retain any aspect of symbolic reasoning, such as those in 1 & 4, we will instead focus on methods that belong to the remaining categories. Furthermore, we are only interested in architectures that integrate both the neural and symbolic components on a fundamental level, and not the act of a model calling an external framework, for example, an LLM calling a symbolic engine, whose output is then transferred back to the LLM, in the midst of a prompt. We establish three main fields of neuro-symbolic AI – Reasoning, Learning and Metacognition, for which we will define and cover some literature.

3 Reasoning

Inaccurate reasoning is a fundamental issue of connectionist AI. Unlike the reasoning process in its symbolic counterpart, which typically calls upon a deductive function, reasoning in neural architectures differs in a way that it is a result of collective passes through internal layers that have been trained to approximate logical results. A classic example of failed reasoning is prompting an LLM with “How many ‘r’s are there in the word ‘strawberry’?”, where the result may be “Two”, contrasting with the fact that three ‘r’s are present. We refer to this phenomenon as *intrinsic hallucination* [9], where statements are verifiable with available reference data, yet incorrect results are still returned. Although commonly associated with LLMs, this phenomenon extends to other DL frameworks such as image recognition or generative AI [10, 14]. The topic of the 2019 Montreal AI Debate was whether a purely connectionist or a hybrid neuro-symbolic framework could better push the boundaries of AI reasoning, driving an increase in research towards hybrid solutions [8]. Among the three fields, Neuro-symbolic Reasoning stands out as the most prominent, focusing on the process of deducing facts through symbolic operations or under symbolic constraints. We cover a subfield of methods that focus on handling logical knowledge and relational data via symbolic constraints, as well as domain/task-specific reasoning approaches.

3.1 Reasoning via Symbolic Constraints

Logic Tensor Networks (LTNs) [4, 27] maps the symbols and rules of first-order logic to the structure of a tensor NN. Continuous, real-valued logic is used to ensure each relation adheres to logical constraints, while remaining differentiable. For example, values remain within the range 0 (false) and 1 (true), while a consistent set of continuous logical operators ($\neg$, $\vee$, $\wedge$) is chosen for the specific task. As weights directly correspond to continuous relations, the forward pass is structurally equivalent to performing deduction in fuzzy logic¹. Also utilizing continuous values, Logic Neural Networks (LNNs) [23] differ from LTNs in that these values serve to recreate absolute Boolean logic through interval semantics instead of fuzzy logic. An intermediate value/neuron, a 2-tuple (L, U) where $L, U \in [0, 1]$, is considered true when $L, U \geq \alpha$, false when $L, U \leq 1 - \alpha$, and uncertain if $L \leq 1 - \alpha$ and $U \geq \alpha$, for a user-defined cutoff $\frac{1}{2} < \alpha \leq 1$. This difference makes LNNs preferred for deductive tasks such as inductive logic programming [26], and LTNs preferred for classification and relational learning. Models based on Markov logic networks (MLNs) [22], a first-order logical framework that assigns a probabilistic distribution across all possible worlds (set of logical rules

¹A continuous logic that permits values between 0 and 1. For example, a truth value of 0.5 may be interpreted as a 50% chance of being true in some contexts. (see also: probabilistic logic)

mapping predicates to truth values), such as pLogicNet [21] and neural MLNs [17] integrate neural components for greater efficiency and flexibility. Neural MLNs replace the explicit first-order rules of MLNs with a single NN, reducing the reliance on manually engineered rules, while pLogicNet combines MLNs with knowledge graph embedding models, for which a neural framework can be chosen, improving reasoning over knowledge graphs. DeepProbLog [15] uses NNs to convert subsymbolic inputs into real-valued logical predicates, which are then transferred into a framework similar to *Prolog* that works on probabilistic logic², where Boolean inference on probabilities takes place. DeepStochLog [34] extends the previous approach by replacing definitive clauses, a set of fixed rules implemented across different tasks, in the logical framework with a neural and stochastic counterpart, enabling tunable and continuous inference, thus providing greater generalizability.

3.2 Domain-specific Reasoning

Multiple notable frameworks successfully deployed by Google DeepMind utilize neuro-symbolic approaches for a diverse set of problems. AlphaGo and its extensions [28, 29], combine Monte-Carlo tree search, an algorithm that processes symbolic game states, with NNs that guide the search through some heuristic. AlphaGo, an engine for *Go*, utilizes both a Policy network to limit the space of possible next moves and a Value network to measure the favorability of a state, which are initially trained with supervised examples before transitioning to RL. AlphaZero builds upon AlphaGo, using only a single NN for both policy and value functions, while full RL with no supervised learning meant it could perform as well for similar games like *chess* and *shogi*. The idea of optimizing game strategy goes beyond simple board games. AlphaTensor [6] applies this idea to find algorithms similar to *Strassen’s algorithm*³ for large matrices. The tensor representation of a standard matrix multiplication forms the initial state, and at each step, a rank-1 tensor is subtracted from the state, corresponding to the result of a single scalar multiplication between some linear combination of matrix entries. Once the goal (zero tensor) is reached, the sequence of steps would represent an alternative formulation of matrix multiplication, where a shorter sequence would lead to a formulation with less scalar multiplications.

AlphaGeometry [32] was also developed for solving equality-based problems in plane geometry (2-dimensional Euclidean geometry), namely those in *International Math Olympiad*. It primarily utilizes a symbolic engine coined Deductive Database + Algebraic Rules (DD+AR). DD is a domain-specific automated theorem prover that deduces new facts based on current ones, which in this case is programmed for plane geometry. AR is an algebraic solver for simple computation on angles, ratios and distances, which may be symbols or numbers without an exact floating-point representation. The neural component of AlphaGeometry does not provide a heuristic to guide the search, instead, it is a language model that proposes auxiliary constructions whenever DD+AR does not reach the goal after some iterations. These auxiliary constructions are not necessary for DD+AR to arrive at the goal state, but instead shorten the process by reducing the number of searches to reach the goal, producing a shorter proof length since it directly corresponds to the goal depth. Notably, AlphaGeometry far surpassed all existing frameworks for a pool of IMO geometry problems, scoring 25/30 as compared to the previous record which only obtained 10/30.

In the domain of conversational language models, Commonsense Reasoning by Instruction (CORGI) [3] pairs a language model with a symbolic engine to better reason with commonsense, or unstated presumptions in human speech. The neural language model is responsible for conversation as well as inferring such presumptions, while the symbolic engine is an automated theorem prover

²A form of logic whose values represent probability distributions between 0 and 1.

³An algorithm for fast matrix multiplication. Strassen’s algorithm between 2×2 matrices results in 7 instead of 8 scalar multiplications.

that attempts to draw conclusions given current knowledge. A neuro-symbolic framework for visual question answering (VQA), ∇ -FOL [1], distinguishes the visual perception and reasoning processes. It maintains neural models for visual perception tasks (scene recognition and subsequently, predicate generation), and a differentiable first-order framework is used for reasoning, in similar fashion to methods previously mentioned.

4 Learning

Shortcut learning [7] is a recurring phenomenon in DL where models often form ‘shortcuts’ or misleading correlations between specific patterns in training data, which could lead to misclassification when generalized. For example, an image captioning model might miscategorize “a green, grassy landscape” as “sheep grazing”, due to the frequent co-occurrence of both features in a dataset. Such issues are hard to diagnose due to knowledge being stored implicitly as weights, without concrete representations. Implicit representations of knowledge also limit the ability of neural models to apply learned knowledge to other contexts [12]. Furthermore, neural models are prone to *catastrophic interference* [18], where previously learned information are overwritten by weight updates to accommodate new knowledge. Symbolic approaches store concrete representations of facts in a knowledge base, which can resolve the aforementioned issues faced by neural models. Here, we define Neuro-symbolic Learning as techniques that focus on the process of representing real-world information (subsymbolic data) as symbolic objects and relations within a model’s knowledge base.

The Neuro-Symbolic Concept Learner (NS-CL) [16] utilizes a three-pronged approach to jointly learn visual concepts, words and semantic language parsing⁴, for VQA. It consists of a neural vision module that converts images to symbolic representations, a neural semantic parser that generates the associated program from the question, which the symbolic unit executes. This framework allows NS-CL to learn directly from images and question-answer pairs, which are baseline training data for VQA tasks, unlike ‘supervised’ approaches that also provide a target program for the semantic parser to learn, achieving record accuracy among other VQA models with no training annotation. Outstandingly, with a 90% reduction of training data, NS-CL displays only a tiny drop in performance (99.1% $\rightarrow$ 98.9%), while the performance of other models degrade significantly.

The integration of knowledge bases and graphs in modern DL frameworks can also serve to provide contextual understanding for specific tasks [20]. For example, automatic driving systems lack basic, commonsense understanding of unusual traffic phenomena not present in training data, and may respond dangerously when encountering such situations. Symbolic knowledge would then be able to provide context as to how the situation, an environment modeled by objects (symbols) within the system, can be better assessed. A *Python* library, DomiKnowS [5] provides a framework to integrate domain-specific symbolic knowledge bases and graphs with deep learning models, improving the accessibility of neuro-symbolic methods for existing DL approaches.

5 Metacognition

The book on human psychology, *Thinking, Fast and Slow*, claims that the human brain switches between two modes of thought: **System 1**, where the thought process is swift and instinctive, the main mode when carrying out things that we are already habituated to; **System 2**, where the thought process is slower and more careful, especially when faced with unfamiliar situations or performing tasks that require precision. Borrowing from these concepts, H. Kautz defines methods within the **Neuro[Symbolic]** category as primarily neural models that are self-aware, switching between Systems 1 (neural) & 2 (symbolic) based on self-assessed confidence on the matter. For

⁴How natural language can be broken down into machine instructions

example, when the neural model is certain that a question is of sufficient simplicity, it can simply assume that its own probabilistic output would be correct. However, when a much harder question is asked, it may deduce that its own parameters are insufficient, and must resort to the slower, but logically precise, symbolic engine, whose output can be used to further refine the neural model. The field of Neuro-symbolic Metacognition, however, is still mostly hypothetical, having little coverage in current literature [24, 31]. Furthermore, while it could be argued that methods in this field may have the potential to surpass general human intelligence, an overarching issue is still the lack of a symbolic System 2 engine that is general enough to address problems across all different fields.

6 Future Directions and Conclusion

While neuro-symbolic approaches in AI have shown great promise in recent advancements, the same must be said for achievements that lie strictly within the field of DL. Neural Module Networks [2] organize knowledge into smaller neural modules, analogous to symbolic knowledge bases, while remaining purely neural, allowing the process of reasoning and knowledge representation in a purely neural system to be made interpretable to some extent. Chain-of-Thought Prompting [33], a modern pipeline for reasoning in LLMs, has also enabled LLMs to scale even greater heights, as seen in *Gemini 2.5 Pro* scoring a gold medal equivalent for the *2025 International Math Olympiad*, with reasoning done end-to-end in natural language. In present day, neuro-symbolic AI still has a long way ahead before achieving universal generalizability. Whether the path forward for AI is neuro-symbolic is unknown — but that is irrelevant — its insights alone may enable even better, accurate and transparent models to be developed in the future.

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